MIL-B-55118A(EL)
14 June 1971

SUPERSEDING
MIL-B-55118(SigC)
10 February 1961

MILITARY SPECIFICATION

BATTERIES, STORAGE, (CELLS), VENTED, NICKEL-CADMIUM

1. SCOPE

1.1 This specification covers portable, vented, rechargeable nickel-cadmium storage batteries, (cells), for use, normally in multi-cell batteries, with military ground equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Film, Polyethylene Thin Gage
L-P-1183	Plastic Molding Material, Acrylonitrile - Butaciene-Styrene (ABS) Rigid
FF-N-836	Nut, Plain, Hexagon, Square, Cap and Welding Nut, Slotted, and Castellated, Hexagon
FF-S-92	Screw, Machine, Slotted or Cross Recessed
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-S-781	Steel Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container grade) and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116	Preservation, Methods of
MIL-B-857	Bolt, Nut and Stud
MIL-M-13231	Marking of Electronic Items
MIL-M-20693	Molding Plastic, Polyamide (Nylon) Rigid

FSC 6140

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MIL-B-43014	Boxes, Water Resistant Paperboard, Folding, Set-up and Metal-stayed
MIL-C-45662	Calibration System Requirements

STANDARDS

FEDERAL

FED-STD-601	Rubber Sampling and Testing

INDUSTRY

USAS1 Y14.5	Dimensioning and Tolerancing for Engineering Drawings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards Order of Precedence for the selection of
MIL-STD-147	Palletized Unit Load, 40 x 48 in, 4 way partial and 4 way pallets
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
MIL-STD-417 (ORD)	Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests)
MIL-STD-454	Standard General Requirements For Electronic Equipment

SPECIFICATION SHEETS

See Supplement - 1 for list of applicable Specification Sheets.

(Copies of specifications, standards, specification sheets and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Military specification sheets for individual battery types. - Detailed requirements or exceptions applicable to individual types are specified on the military specification sheets of the specification. In the event of any conflict between requirements of this specification and the individual military specification sheet the latter shall govern. Dimensioning and tolerancing shall meet requirements of Standardization Document USAS1 Y14.5.

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3.2 Classification of requirements. -

Requirement	Paragraph
Selection of specification and standard	3.3
First article	3.4
Materials and components	3.5
Design and construction	3.6
Potassium carbonate	3.7
Attitude	3.8
Electrical leakage	3.9
Mechanical shock	3.10
Vibration	3.11
Temperature cycling	3.12
Charged storage	3.13
Cycle life	3.14
Capacity	3.15
Labeling and marking	3.16
Battery condition for shipping	3.17
Workmanship	3.18

3.3 Selection of specifications and standards. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 First Article. - Cells furnished under this specification shall be a product which has been subjected to, and passed the first article inspection specified herein.

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3.5 Materials and components. - When a definite material or component is specified, it shall be in accordance with the applicable specification or requirement listed in Table I. When deemed necessary by the Government, certification from the source of the material or component will be required. In the absence of certification from the source, a certificate of analysis or certified inspection data will be required. (See 4.3)

TABLE I - Materials and Components

Material or Components	Applicable Specification or requirements, (See 4.3)	Method of Test
Metals	3.6.1	4.7.5
Nickel plating	QQ-N-290	
Nuts	FF-N-836	
Machine screws	FF-S-92	
Bolts and studs	MIL-B-857	
Plastic, Molded (Container)	L-P-1183, MIL-M-20693	4.7.2 and 4.7.4
Elastomers (Vent gaskets or O rings and sleeves)	MIL-STD-417 (Grd)	4.7.6 FED-STD-601
Filler caps	3.6.3	4.7.8

3.6 Design and construction. -

3.6.1 Metals. - All metals which do not enter into the basic electrochemical reaction of the cell shall resist or be treated to resist corrosion. Metal parts shall show no evidence of pitting, scaling, corrosion or other deleterious effects after being tested, as specified in 4.7.5 and elsewhere herein.

3.6.1.1 Dissimilar metals. - When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.6.2 Container. - The plastic cell container shall maintain its required dimensions throughout inspection. Cells may be restrained to simulate assembly into a battery assembly to meet this requirement. The container shall be made of polyamide (nylon) per MIL-M-20693, acrylonitrile, butadiene - styrene (ABS), per L-P-1183 or equal, with sufficient thickness and rigidity to withstand deformation. The plastic material shall not change more than 2.5% in weight or 2% in dimensions nor shall it blister, crack or pit when tested per 4.7.4. All surfaces shall be smooth, free from pitting, blow holes, rough spots, or other deformation. Container shall not leak, be distorted or rupture when subjected to internal pressure test. (See 4.7.2)

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3.6.3 Venting system. - Each cell shall be equipped with a bayonet type filler cap per applicable specification sheet complete with elastomer gasket or O ring and vent sleeve. Vent body shall be polyamide per MIL-M-20693 ABS or equal. Filler cap shall release gas within limits specified in applicable specification sheet when tested per 4.7.8.

3.6.4 Terminals. - Each cell shall be equipped with two terminals, either stud and nut or internal thread with slotted hex. hd. screw per 3.1. If screw type, a washer and Belleville spring shall be furnished on each terminal. If stud and nut type, each stud shall be furnished with two nuts and Belleville spring. Electrical contact surfaces shall not be contaminated by insulating substances.

3.6.4.1 Terminal seal. - The seals between the terminals and cell tops shall be constructed so that the cell is capable of meeting all test requirements herein without evidence of electrical leakage as determined by 4.7.9.

3.6.5 Elastomer material. - All elastomer materials used to fabricate O-rings, gaskets, sleeves and pads shall have a permanent set of less than 6% and shall show no signs of deterioration after being subjected to test specified in 4.7.6.

3.6.6 Electrolyte. - Electrolyte shall be approximately 1.300 specific gravity potassium hydroxide (KOH).

3.6.7 Dimensions and weights. - The dimensions and weights of the cells shall be as shown on the applicable specification sheets.

3.7 Potassium carbonate. - After one charge and discharge, the potassium carbonate (K_2CO_3) content of the electrolyte, as determined in 4.7.7, shall not exceed 60 grams per liter in high rate cells. (See 3.1)

3.8 Attitude. - The cell shall meet the electrical capacity requirements in any specified position. (See 4.7.3, 4.7.14.3, Tables III and V)

3.9 Electrical leakage. - When the cell is tested in accordance with 4.7.9, the electrical resistance shall be more than 50 megohms.

3.10 Mechanical shock. - The cell shall meet the mechanical shock, (4.7.11) requirements of Military Standard MIL-STD-202 as follows:

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TABLE II

Cells Weighing 4 Pounds or Less	Cells Weighing More than 4 Pounds
Test Method 202B at a maximum of 50 gravity units, (g's)	Test Method 205C at a maximum of 50 gravity units, (g's)

Capacity shall be met per 4.7.14.2 and 4.7.14.3 after mechanical shock.

3.11 Vibration. - The cell shall meet the low frequency vibration requirements of Method 201 Military Standard MIL-STD-202, as modified in 4.7.12. Capacity shall be met per 4.7.14.2 and 4.7.14.3 after vibration.

3.12 Temperature Cycling. - The cell shall not be detrimentally effected, such as cracked or ruptured case, when subjected to temperature cycling in 4.7.10. Capacity shall be met per 4.7.14.2 and 4.7.14.3 after their exposure.

3.13 Retention of charge. -

3.13.1 Normal storage retention of charge. - The cell shall furnish capacity specified in applicable specification sheet after conditioning of 4.7.14.1, if required; charge per 4.7.14.2 and storage under normal conditions of 4.6.1 and 4.7.13.1.

3.13.2 High temperature storage retention of charge. - The cell shall furnish capacity specified in applicable specification sheet after conditioning of 4.7.14.1, if required; charge per 4.7.14.2, and after storage @ 160 $\pm$ 4°F per 4.7.13.2.

3.14 Cycle life. - The cell shall meet the cycle life requirements of the applicable specification sheet and 4.7.15.

3.15 Capacity. - Cells shall provide the required service for the applicable capacity tests of 4.7.14 when discharged in accordance with specified load, final voltage and temperature conditions. (See 3.1)

3.16 Labeling and marking. - All labeling and marking shall be clear and legible throughout all of the tests specified herein. The lettering shall be black in color and hot stamped or equal in accordance with MIL-M-13231.

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3.16.1 Identification label. - The identification label shall consist of the following:

Minimum Height (In.)	Marking
3/16	BATTERY, STORAGE
3/16	BB-__*__/U
1/16	(FED STOCK NO) 6140-__*__
1/16	(MFR'S NAME) __*__
1/16	(CONTRACT NO.) __*__
1/16	(DATE OF MFR'S CODE) __*__
1/16	1.2V, _____A.H. @ _____
1/16	_____DATE PLACED IN SERVICE
3/16	US

* Applicable information shall be marked on cell by manufacturer.

Delete information shown inside parentheses.

Example:

BATTERY, STORAGE
BB-418/U
6140-855-7634
ART DUZE BATTERY CO.
DAAB05-68-C-0000
0568
1.2 V, 14 A.H. @ 5 HR RATE
_____DATE PLACED IN SERVICE
US

3.16.2 Month of manufacture code. - The code shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the month and the last two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0". When a battery is completed during the last three working days of a month or the first three working days of the following month, the manufacturer may use either of these months as the code date of manufacture.

Examples:

A battery manufactured in November 1968 will bear the code "1168".

A battery manufactured on 29 May 1969 will bear the code "0569" or "0669".

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3.16.3 Terminal marking. - Positive or both terminals shall have proper polarity identification. Marking shall be apparent and permanent. It shall appear on top of terminal or cell. When on cell top, it shall be readily visible and as close as feasible to the proper terminal. Marking on cell top shall be red for positive and black for negative. Supplemental color coding on terminals is optional but electrical contact portions shall not be covered with paint or other insulating materials. The following polarity identifications are acceptable:

POSITIVE	NEGATIVE
POS, P or +	NEG, N or -

3.16.4 Electrolyte level measurement. - Cells shall be constructed so that electrolyte levels above top of plates can be determined by means of a hollow glass rod.

3.17 Cell condition for shipping. - Unless otherwise specified, each cell shall be furnished completely discharged; properly filled with electrolyte (3.6.6) so that ultimate capacity is attained without spewing of electrolyte during charge; complete with one or two piece vent cap (3.1); and with dust cap inserted. A warning tag instructing the user to remove the dust cap when charging shall be included in the intermediate package.

3.18 Workmanship. - Cells shall be processed in such a manner as to be uniform in quality and free from defects that will affect their life, serviceability, or appearance. The criteria for acceptable workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - The inspection requirements specified herein are classified as follows:

(a) Materials and components inspection. (See 4.3)

(b) First Article inspection. (See 4.4)

(c) Quality conformance inspection.

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(1) Inspection of product before preparation for delivery. (See 4.5.1)

(2) Inspection of preparation for delivery. (See 4.5.2)

4.3 Materials and components inspection. - Materials and components inspection shall consist of verification by certification from the battery (cell) supplier to authorized Government inplant representative that the materials and components utilized in the fabrication of the cells are in accordance with Table I.(See 3.5) Certification shall be furnished by supplier prior to first article approval and upon request of Government during production. Certification shall indicate that the component supplier or cell supplier have met all requirements of Table I. In the absence of certification from the source, a certification of analysis or certified inspection data shall be required as proof of conformance to applicable requirements by the cell supplier.

4.3.1 Samples of materials and components. - For those items listed in Table I for which the specification requirements do not reference a subsidiary specification, eight samples of materials or components, treated and processed as they would be in the finished battery, shall be inspected as covered herein.

4.4 First article inspection. - It shall consist of the examination and tests listed in Table III and shall be performed in the order shown therein. Where first samples of production in lieu of preproduction samples are selected at random for inspection, Group A and B quality conformance inspection need not be performed for the first month's production.

4.4.1 Sample cells. - The supplier at no extra cost to the Government shall make twelve (12) sample cells, which will constitute a first article inspection lot.

4.4.2 Failure. - If one or more sample cells fail any of the first article inspections, the supplier shall immediately determine cause of failure, take the necessary remedial action and subject a second lot of first article samples to all the inspections in Table III. Official approval will be given only when an entire lot of first article sample batteries have passed all required examinations and tests. If failure is deemed minor by Procuring Contracting Officer, a complete new set of test samples need not be furnished and retested. However, the number of new samples and amount of testing shall be commensurate with the degree and type of failure.

4.4.3 Start of production. - Any production prior to approval of certification of materials and components and first article inspection of Table III shall be at the supplier's own risk.

4.5 Quality conformance inspection. -

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4.5.1 Quality conformance inspection of product before preparation for delivery. - The contractor shall perform the inspection specified for Groups A, B, and C. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4) Test Equipment for Government verification inspection shall be made available by the contractor.

TABLE III - First Article Inspection

Inspections	Sample Numbers	Requirement Paragraph	Inspection Paragraph
Visual and mechanical examination	1 to 12	3.6.1.1, 3.6.4, 3.6.7, 3.16 thru 3.16.4, 3.17 and 3.18	4.7.1
Initial Full capacity			
Discharged (upright)	1 to 6	3.15	4.7.14.1
Discharged (on side)	7 to 12	3.15, 3.8	4.7.3
Cycle life	1 to 4	3.14	4.7.15
Potassium carbonate	5 to 8	3.7	4.7.7
Retention of charge I (14 days)	5 to 8	3.13.1	4.7.13.1
Retention of charge II (4 days)	9 to 12	3.13.2	4.7.13.2
High rate discharge	9 to 12	3.15	4.7.14.4
Low temperature	9 to 12	3.15	4.7.14.5
Temperature cycling	9 to 12	3.12	4.7.10
Full capacity (upright)	9 to 12	3.15	4.7.14.3
Mechanical shock	9 to 12	3.10	4.7.11
Full capacity (upright)	9 to 12	3.15	4.7.14.3
Vibration	9 to 12	3.11	4.7.12
Full capacity (upright)	9 to 12	3.15	4.7.14.3
Internal pressure	1 to 12	3.6.2	4.7.2
Electrical leakage	1 to 12	3.9	4.7.9

4.5.1.1 Group A inspection. - Each cell of each lot shall be inspected per Table IV for conformance to applicable inspection requirements of Table VIII. Each lot will be subject to verification, utilizing the general inspection level II of Military Standard MIL-STD-105, and the acceptable quality levels (AQL's) indicated in Table IV. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE IV.- Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL(%) Major	AQL(%) Minor
Visual and Mechanical (External)	3.6.1.1, 3.6.4, 3.16 thru 3.16.4, 3.17 and 3.18	4.7.1	1.0	4.0
Internal pressure	3.6.2	4.7.2	1.0	*

* All defects are considered major.

4.5.1.1.1 Disposition of samples. - The sample cells, having been subjected to and passed Group A inspection shall be allocated to Group B inspection as required. Group A sample cells not allocated for Group B inspection shall be retained as part of the lot until disposition is determined by Group B inspection.

4.5.1.2 Group B inspection. - Group B inspection shall consist of tests shown in Table V and shall be performed in the order listed, unless otherwise specified. Statistical sampling shall be in accordance with special inspection level S-3 of MIL-STD-105 and the AQL specified for each test in Table V.

TABLE V - Group B Inspection

Test	Requirement Paragraph	Inspection Paragraph	AQL
Initial full capacity (50% upright & 50% on side)	3.15, 3.8	4.7.3 and 4.7.14.1	4.0%
Potassium carbonate	3.7	4.7.7	2.5%
Retention of charge (4 days)	3.13.2	4.7.13.2	4.0%
High rate discharge	3.15	4.7.14.4	4.0%
Mechanical shock	3.10	4.7.11	4.0%
Full capacity	3.15	4.7.14.3	4.0%
Electrical leakage	3.9	4.7.9	2.5%
Internal pressure	3.6.2	4.7.2	2.5%

4.5.1.2.1 Disposition of samples. - Group B sample cells shall not be delivered under the contract against the quantity on order. At the option of the supplier, these samples may be selected in accordance with 4.5.1.3 for Group C inspection.

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4.5.1.3 Group C inspection. - The Group C inspection shall consist of the tests listed in Table VII and they shall be performed in the order shown for each respective subgroup. The release of shipment lots, which have passed Group A and B inspection, shall not be held up pending results of Group C testing performed on sample batteries representative of these lots. The number of samples required for Group C inspection shall be in accordance with Table VI and shall be samples that have passed Group A and B Inspection. The cells shall be selected randomly as soon as possible in such a manner as to be representative of the entire contract lot. (Group C is not performed on first lot if first samples of production are inspected. Group C is not necessary if first samples of production are required and only one lot is produced.)

TABLE VI - Group C Sampling Plan

Contract Lot Size	No. of Samples		
	Subgroup I	Subgroup II	Total
Up to 50	1	1	2
51 to 280	2	1	3
281 to 1200	3	2	5
1201 to 3200	5	3	8
3201 to 10,000	8	5	13
10,000 to 35,000	12	8	20
35,001 to over	20	12	32

TABLE VII - Group C Inspection

Inspection Type	Subgroup I	Subgroup II	Req.Par.	Insp.. Para.
Retention of charge (14 days)	X		3.13.1	4.7.13.1
Cycle life		X	3.14	4.7.15
Temperature cycling	X		3.12	4.7.10
Full capacity (upright)	X		3.15	4.6.14.3
Low Temperature discharge	X		3.15	4.7.14.5
Vibration	X		3.11	4.7.12
Full capacity	X		3.15	4.7.14.3
Internal pressure	X	X	3.6.2	4.7.2
Electrical leakage	X	X	3.9	4.7.9

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4.5.1.3.1 Disposition of samples. - Group C sample cells shall not be delivered under the contract against the quantity on order.

4.5.1.3.2 Non-compliance. - Failure of a Group C test sample to pass any specified test shall be cause for the Government to refuse to accept subsequent lots of the type under consideration until it has been proved to the satisfaction of the Government that the cause of failure revealed by testing has been eliminated. The contractor shall report immediately in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall investigate immediately the cause of failure and further report the results of investigation and details of proposed corrective action on (a) the process and materials, as applicable, and (b) all batteries which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample batteries shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.2 Quality conformance inspection of preparation for delivery. - Inspection of preparation for delivery shall be in accordance with requirements as specified in Specification MIL-P-116 to verify conformance with requirements of section 5 herein.

4.6 Inspection conditions and equipment. -

4.6.1 Standard test conditions. - Unless otherwise specified herein, all measurements and inspection shall be conducted as follows:

Variable	Limits
a. Temperature	80 ± 20°F (not capacity & cycle life tests) 80 ± 10°F (capacity and cycle tests)
b. Humidity	Uncontrolled
c. Pressure	Uncontrolled
d. Storage time between charge and discharge (except cycle life test or otherwise specified)	Minimum of 30 minutes

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Variable (Cont'd)	Limits (Cont'd)
e. Predischarge storage conditions where temperature prior to test is other than that temp.	Minimum of 24 hours @ test temp.; or 16 hours and center cell thermocouple indication that sample temp. is within specified limits. Maximum of 60 hours.

4.6.1.1 Low temperature test conditions. - Unless otherwise specified, low temperature tests shall be conducted in accordance with foregoing Paragraph 4.6.1 except the temperature shall be -40 ± 2°F.

4.6.2 Test equipment and inspection facilities. - Contractor shall provide adequate test equipment and inspection facilities to assure accurate test data records, without delay, of all inspection required herein. The contractor shall establish and maintain a system for the calibration of all measuring and test equipment in accordance with Military Specification MIL-C-45662.

4.6.2.1 Meters. - All voltmeters and ammeters shall be accurate within ± 1 percent of the full scale reading. Meter ranges shall be such that all readings are taken on the upper half of the scale. Sensitivity of voltmeters shall be at least 100 ohms per volt.

4.6.2.2 Pressure equipment. - The accuracy of the manometers and gages shall be ± 10% for all measurements between 0 and 25 psi.

4.6.2.3 Timing equipment. - Timing equipments shall be accurate within ± 0.5 percent.

4.6.2.4 Test loads. - Constant current test loads shall be adequately variable and controllable to assure that current drains or rates specified are maintained within ± 1 percent. Wattage ratings of inspection equipment shall be sufficiently high so that test equipment is not overloaded. Load resistance for fixed time discharges shall be determined by $R_L = 5.2/C_5$ (5 hr rate capacity). Resistance tolerance of load resistances used for fixed time discharges of life test shall be ± 5%, (this shall include entire load circuit).

4.7 Methods of examination and tests. -

4.7.1 Visual and mechanical examination. - Batteries submitted for acceptance under contract shall be examined to determine compliance with specification requirements as to workmanship, design, construction, dimensions, weight, labeling and marking. The classification of defects shall be in accordance with Table VIII.

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TABLE VIII Classification of Visual and Mechanical Defects

Categories	Defects
Major	
101	Wrong parts or material.
102	Improper assembly causing parts to be inoperative or unsafe in service.
103	Deformed or damaged parts, which are inoperative or malfunction in service.
104	Dimensions out of tolerance which affect interchangeability.
105	Maximum allowable weight exceeded.
106	Insulating parts or materials missing, damaged, or improperly located so as to affect electrical performance.
107	Location, polarity and marking of terminals not as specified.
108	Labeling and marking wrong, missing or illegible.
109	External and internal threads missing, wrong size or so damaged to prevent proper use.
110	Sealing missing or defective resulting in electrolyte leakage.
111	Welded or soldered connections improperly made so as to adversely affect battery performance.
Minor	
201	Improper assembly which could reduce efficiency of operation but not render cell inoperative or unsafe in service.

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TABLE VIII Classification of Visual and Mechanical Defects. (Cont'd)

Categories	Defects
Minor	
202	Deformed or damaged parts which do not affect performance.
203	Inferior insulating parts or materials which do not adversely affect electrical performance.
204	Dimensions out of tolerance which would not affect interchangeability or operation.
205	Improper marking which doesn't hamper utilization or identification of the cell.

4.7.2 Internal pressure test. - The cell may be restrained in a test case. An airtight adapter shall be fitted into the filler cap opening of the cell, with a gauge and controllable air pressure up to 25 psi. Standard test conditions of 4.6.1 a, b and c apply. Upon completion of air tightness test of test equipment and with the cell completely immersed in water, the air pressure within the cell shall be slowly adjusted to 20 psi and maintained at that pressure for 30 seconds. The emission of air bubbles from cell during this period will indicate a failure due to leakage. The cell shall be measured to determine if it is still within specified tolerance, upon completion of pressure phase of this test.

4.7.3 Attitude. - The cells shall be tested in position specified herein.

4.7.4 Alkali absorption. - Three 2 x 2 in. specimens shall be cut from each plastic material submitted by the contractor. The cut edges may be sealed with the same resin used in fabrication of the cell component. After making accurate initial measurements of all dimensions with device such as vernier caliper, the specimens shall be weighed accurately by means of an analytical balance. Weights and dimensions shall be recorded complete with corresponding specimen number or identification. Measurements shall be taken at an ambient of 68 to 80°F and maximum of 80% relative humidity. The specimens shall be immersed individually so that all but one side shall be completely exposed to the potassium hydroxide solution. The test vessels shall be covered and contain 150cc of 1.300 ± 0.020 specific gravity potassium hydroxide. These vessels shall be held for 7 days in an oven @ 150°F ± 4°F. At the end of the seven days, the specimens shall be removed immediately from the oven, rinsed carefully and then dried on the surface. The three specimens shall then be measured, weighed and visually inspected to determine if requirements of 3.6.2 have been met by the plastic.

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4.7.5 Corrosion resistance. - Eight samples of all external metal parts of cell shall comply with 3.6.1 after being subjected to the following tests:

Test I. - Salt spray test in accordance with Method 101B of MIL-STD-202 for a continuous period of 100 hours.

Test II. - Wet one-half of each specimen with a solution of 1.300 ± 0.020 specific gravity potassium hydroxide and allow to stand for 48 hours. Specimens shall then be rinsed, dried and observed.

4.7.6 Alkali resistance. - Three samples of gaskets or O rings, made of elastomer material, shall be cleaned, rinsed and immersed for 30 ± 1 days in an aqueous solution of 1.300 ± 0.020 specific gravity potassium hydroxide under normal test conditions of 4.6.1. Samples shall then be removed, rinsed with water and allowed to dry. They shall then be subjected to Test Method 3311 of Federal Standard 601.

4.7.7 Potassium carbonate. - Sufficient electrolyte shall be removed from charged sample cells, after they have been subjected to one charge and discharge, for the purpose of analysis. Each potassium hydroxide sample shall be subjected to the following standard titration method:

(a) Determine specific gravity and temperature of electrolyte.

(b) Extract 1 mm electrolyte from cell with clean, dry pipette.

(c) Transfer to beaker (250 ml) and dilute with carbon dioxide free distilled water (bubble CO_2 free air through distilled water for a minimum of 15 hours).

(d) Add 3-5 drops phenolphthalein per liter of 95% alchohol.

(e) Titrate with 1 N soln. HCL or H_2SO_4 to end-point which is the disappearance of the pink color.

(f) Add a few drops of methyl violet to solution and again titrate to end point (violet to grey to green).

(g) Record the titers as follows:

A = Phenolphthalein end-point
B = Total titration therefore:
B-A = b.
Value of B = Normality of total alkalinity
A = potassium hydroxide equivalent
b = potassium carbonate (K_2CO_3) equivalent
The value of "b" (K_2CO_3) shall not exceed that specified in 3.7.

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4.7.8 Venting. -

4.7.8.1 Normal temperature venting. - Under normal test conditions of 4.6.1 the sample filler cap shall be connected to an adaptor to which a variable pressure source of compressed air is connected. The filler cap, with adaptor, shall be totally immersed in water and the air pressure, shown by gage or manometer, shall be varied from 1/2 psig below the lower limit and 1/2 psig above the upper limit (3.1). If there is an escape of air bubbles from the vent sleeve between specified limits, the sample has not these requirements. (3.6.3)

4.7.8.2 Low temperature venting. - The low temperature venting limit test shall be performed in accordance with 4.7.8.1, except immersion fluid shall be an inert and non-corrosive solution that has been precooled to -40 ± 2°F and filler cap precooled a minimum of 48 hours @ -40 ± 2°F.

4.7.9 Electrical leakage. - Each cell shall be tested for electrical leakage under standard conditions of 4.6.1.

(a) Charge the cell completely per 4.7.14.2.

(b) Adjust the electrolyte to the proper level.

(c) Tighten filler caps.

(d) Thoroughly scrub the outside surfaces of the cell with detergent, and brush with non-metallic bristles.

(e) Rinse the cell thoroughly with tap water.

(f) Dry completely.

(g) Invert cell for two minutes.

(h) Set cell up-right.

(i) Immediately perform one of the following tests:

TEST I. If the cover-to-cell container seal is less than 1/8 inch from the top of the cell, attach one probe of an ohmmeter to one of the cell terminals and the other to a completely dampened piece of absorbant cotton. The probe with moist cotton shall be moved along the cover-to-cell container line from point of initiation around the cell and back to starting point, around vent and terminal seals, without touching the terminal.

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TEST II. If the seal between the cover and the container is 1/8 inch or more from the top of the container, immerse the cell to a level slightly over the cell-to-container line so that no water comes in contact with top of cover or terminals. After immersion for a minimum of fifteen seconds, a reading shall be taken on an ohmmeter between the water and one of the terminals. Take readings around vent and terminals to a terminal as outlined in foregoing Test I.

4.7.10 Temperature cycling. - With the cell in a discharged condition, it shall be subjected to temperature cycling, Method 102, Test Condition D of MIL-STD-202. The container shall be examined for breaks, cracks or other defects resulting from this storage. Samples shall then meet the capacity requirements of Para. 3.15.

4.7.11 Mechanical shock. - The cell shall meet the rest requirements of 3.10 under normal test conditions of 4.6.1 in a completely discharged condition. The cell shall be examined for defects resulting from exposure to this shock. Samples shall then meet capacity requirements of Para. 3.15.

4.7.12 Vibration. - The cell shall meet the test requirements of 3.11 with the following exceptions, under normal test conditions of 4.6.1 and in a completely discharged condition:

(a) The frequency shall be varied either uniformly or logarithmically between the limits of 10 and 55 cycles per second.

(b) The length of time in each direction shall be 95 $\pm$ 5 minutes for a total of 285 $\pm$ 15 minutes for the complete exposure to low frequency vibration.

(c) The samples shall then be examined for defects due to vibration.

(d) The samples shall then be subjected to and meet capacity requirements of Para. 3.15.

4.7.13 Retention of charge. - The cells shall be given a normal charge in accordance with 4.7.14.2 and shall meet the reduced capacity requirements of the applicable 3.1 after the following storage conditions.

4.7.13.1 Normal temperature retention of charge. - The cells shall be stored under normal conditions of 4.6.1 for 14 days and then discharged under normal conditions of 4.6.1 and applicable specification sheet.

4.7.13.2 High temperature retention of charge. - The cells shall be charged; stored for four days ($\pm$ 2 hours) under conditions of 4.6.1 except 160 $\pm$ 4°F in lieu of 80 $\pm$ 20°F; and discharged per 4.7.13.1.

4.7.14 Capacity. - All capacity tests shall consist of a charge and discharge.

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4.7.14.1 Initial full capacity. - This test shall consist of one to three test cycles at normal test conditions of 4.6.1 and discharges per 3.1. Cells shall be charged per 4.7.14.2 prior to discharge.

4.7.14.2 Normal charge. - The conditions during charge after each discharge shall be normal in accordance with 4.6.1 and applicable spec. sheet, except the low temperature test and portions of the life cycle test.

4.7.14.3 Full capacity. - Cells to be subjected to full capacity discharge shall be discharged under normal conditions of 4.6.1 and 3.1. Cells shall be charged per 4.7.14.2 prior to discharge.

4.7.14.4 High rate discharge. - The high rate discharge (3.0 C_5) is to be performed on cells designed for high rate applications per para. 3.1 and under normal conditions of 4.6.1. Cells shall be charged per 4.7.14.2 prior to discharge. The terminal voltage shall be above 1.15 volts after 5 seconds of discharge and meet capacity of applicable spec. sheet. Cells may be physically restricted to simulate the restriction when installed in a battery.

4.7.14.5 Low temperature. - Both the charge and the discharge shall be conducted @ -40 ± 2°F, under conditions of 4.6.1.1. Sample shall be precooled to -40 ± 2°F prior to both charge and discharge. If temperature of electrolyte is not monitored during inspection, a minimum of 24 hours exposure @ -40 ± 2°F is required before initiation of charge or discharge is allowed. The charge @ -40°F shall be an option of constant current charge in accordance with normal charge except terminate starting current @ 1.68 volts per cell; or constant potential @ 1.68 volts per cell for eight (8) hours or until the charge current is below one (1) amp. for 1/2 hour. Voltages shall be taken during discharge (.2 C_5) until the voltage falls below 1.0 volt. During and after prestorage, following removal from -40°F test facility, outside of cells shall be dried, if required, and visually inspected for electrolyte leakage or deterioration of cell components.

4.7.15 Cycle life. - The cycle life test shall be conducted under normal test conditions of 4.6.1 a, b, and c.

(a) Cycle 1. - The cells shall be charged in accordance with 4.7.14.2 and then discharged @ 0.23 C_5 for 2 hours and 40 minutes.

(b) Cycles 2 thru 109. - The cells shall be charged for 3.33 hours @ 0.23 C_5 amps and immediately discharged for 2.67 @ 0.23 C_5 amps. (See test load requirements in 4.6.2.4)

(c) Cycle 110. - The cells shall be charged for 3.33 hrs @ 0.23 C_5 and immediately discharged @ 0.2 C_5 to an end voltage of 1.0 volt.

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(d) Cycles 111 and 112. - The cells shall be charged in accordance with 4.7.14.2 and then discharged @ 0.2 C_5 to an end voltage of 1.0 volt.

(e) Internal pressure test shall be conducted in accordance with 4.7.2.

(f) Repeat the foregoing, "a" through "e" until end of test (see "g" below).

(g) End of test shall be governed by one of the following conditions:

(1) Cell electrolyte leakage, spewing, foaming, or failure to meet internal pressure test.

(2) Cell doesn't meet capacity of specification sheet on cycle 112 or multiple thereof.

(3) Passed the number of cycles specified on applicable specification sheet.

(h) A minimum of three intermediate cycles shall be performed per day, five days per week.

(i) If capacity failure is encountered during life test, a check test consisting of shorting for 16 hours and conducting full charge capacity should be included.

5. PREPARATION FOR DELIVERY. -

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified. (See 6.2c)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each cell shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each cell shall be dried with applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Each cell shall be individually packaged in accordance with Method 1C-2 of MIL-P-116. Each cell shall be placed in a close-fitting box conforming to MIL-B-43014, Form 1, Style II, Type A, Class A. Box closure shall be in accordance with the appendix of the box specification. Place each boxed cell in a bag, fabricated of material conforming to L-P-378, type 1, grade B, finish 1, having a uniform thickness of 4.0 $\pm$ 0.5 mils. The bag closure shall be by heat seal.

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5.1.1.4.1 Intermediate packaging. - A quantity of cells, bearing the same stock number, packaged as specified in 5.1.1.4, shall be placed in close fitting box conforming to MIL-B-43014, Form 1, Style II, Type A, Class A, not to exceed the weight limitation of 20 pounds. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - Cells shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. -

5.2.1.1 Consolidation. - A quantity of cells, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

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5.2.2 Level B. -

5.2.2.1 Consolidation. - A quantity of cells, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament type conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of cells, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Battery, storage (cell) vented, portable, nickel-cadmium is used to furnish d.c. power for electronic and communications equipment, and other ground power requirements.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required, (BB-_____/U and FSN 6140 _____).

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C)

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(d) Name and address of responsible Government technical activities.

6.3 Definitions. -

6.3.1 Current rate. - Current rate is rate of discharge or charge. The current rate is defined as:

$I = XCy$

I = Current rate in amperes

X = Rate factor in hours

C = Ampere-hour capacity rating of cell

Y = Hourly rate, such as 5 hour rate etc.

For example, $0.2C_5$ indicates a current rate of 0.2(25) or 5.0 amperes for a 25 ampere-hour battery rated at the 5 hour rate.

6.3.2 Cycle. - Temperature cycle is a series of temperatures that cells are exposed to, for the purpose of testing materials to ascertain their resistance to rapid and extreme temperature variations. Capacity or life cycle is a combination of a charge and its subsequent discharge.

6.3.3 Discharged cells. - Cells shall be considered discharged when a specified end voltage has been reached.

6.3.4 Final voltage. - The final, minimum or end voltage is the minimum allowable voltage during discharge. It is specified as final voltage in the specification sheets.

6.3.5 Fully charged cells. - Cells shall be considered fully charged when the charge is in accordance with normal charge of 4.7.14.2.

6.3.6 Fully discharged cells. - Cells shall be considered fully discharged when discharged to zero volts.

6.3.7 Measured performance. - The actual numerical figure obtained in conducting a test in accordance with Section 4 of this specification, be it discharge voltage, current, capacity, cycle life, storage life or anything else, shall be known as the measured performance.

6.3.8 Specified performance. - The actual numerical figure given in the spec. sheet as a performance requirement, be it nominal voltage, end voltage, capacity, cycle life, storage life or anything else, shall be known as the specified performance.

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6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Approval to ship. - Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of a corrective action.

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